# Project Design Phase-II
# Data Flow Diagram & User Stories

| Field | Details |
|---|---|
| **Project Design Phase** | **Phase-II** |
| **Project Name** | **Serene-Study: AI-Based Exam Anxiety Detector** |
| **Date** | **March 7, 2026** |
| **Team ID** | **GACSE-109** |
| **Maximum Marks** | **4 Marks** |

# 1. Data Flow Diagrams

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

### Example: (Simplified) Flow

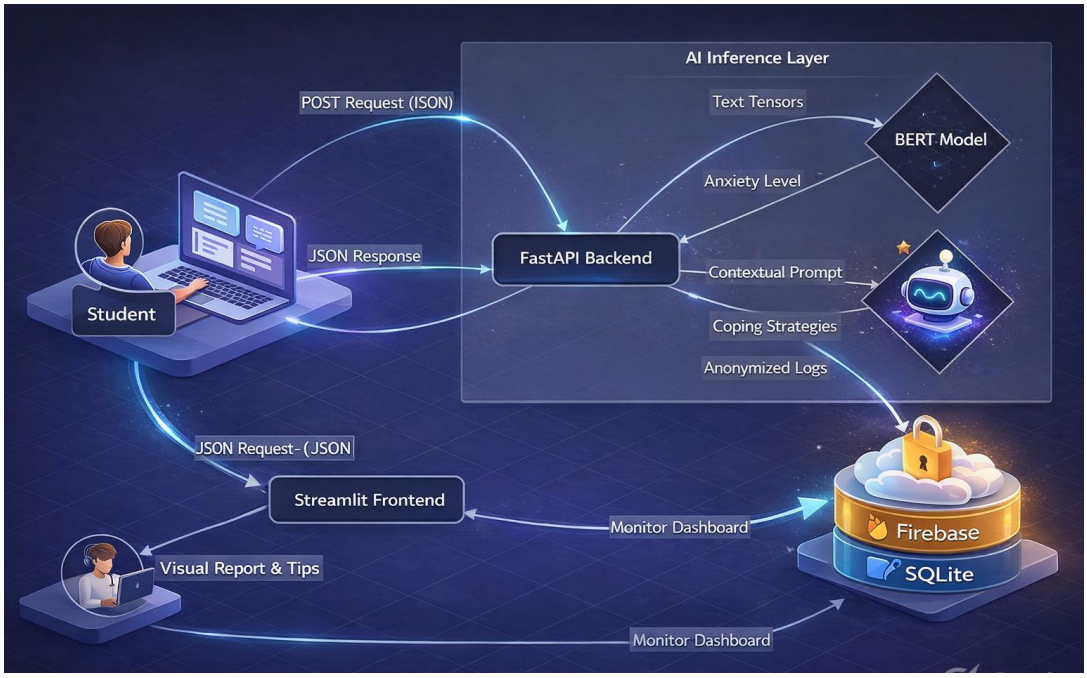
1. **User Input:** The student enters exam-related thoughts and selects their MBTI personality type in the Streamlit web interface.
2. **Request Dispatch:** The frontend captures the input and dispatches a JSON-encoded POST request to the FastAPI /predict endpoint.
3. **AI Classification:** The backend tokenizes the text and executes the fine-tuned **BERT** model to classify the input into an anxiety level (Low, Moderate, or High).
4. **Contextual Enrichment:** The classified level and MBTI type are passed to the **Gemini AI Agent**, which generates personalized coping strategies.
5. **Output & Persistence:** Enriched data is visualized in the UI with Plotly charts and management tips, while the transaction is logged to **Firebase Firestore**.

### Example: DFD Level 0 (Industry Standard)

### System Flow (Crisp Points)

1. **Student Input**
   **Student enters exam thoughts in the Streamlit frontend.**
2. **API Request**
   **Frontend sends a POST request (JSON) to the FastAPI backend.**
3. **Text Processing**
   **Backend sends the text to the BERT model.**

4. **Anxiety Detection**
   **BERT analyzes the text and predicts the anxiety level.**
5. **Prompt Creation**
   **FastAPI builds a contextual prompt based on the anxiety level.**
6. **AI Advice Generation**
   **Gemini AI generates coping strategies and wellness advice.**
7. **Data Storage**
   **Interaction logs are stored in Firebase Firestore and optionally SQLite.**
8. **Response to Frontend**
   **Backend sends results back to Streamlit as JSON response.**
9. **Result Display**
   **Streamlit shows the visual report and coping tips to the student.**
10. **Monitoring**
    **Aggregated data helps counselors monitor anxiety trends.**



# 2. User Stories

Use the below template to list all the user stories for the product.

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| **Student (Web User)** | **Anxiety Assessment** | USN-1 | As a student, I want to enter my feelings about exams so I can understand my stress level. | System returns a Low/Moderate/High classification. | High | Sprint-1 |
| **Student (Web User)** | **Personalized Coaching** | USN-2 | As a user, I want to select my MBTI type so that the advice I receive is tailored to my personality. | Coping tips align with the selected MBTI traits. | High | Sprint-1 |
| **Student (Web User)** | **Interactive Support** | USN-3 | As a user, I want to receive immediate breathing exercises and tips when my anxiety is high. | UI displays actionable tips immediately after analysis. | Medium | Sprint-1 |

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| **Administrator (Counselor)** | **Trend Dashboard** | USN-4 | As a counselor, I want to see a chart of overall student stress levels across the campus. | Dashboard displays an aggregate Plotly chart of results. | Medium | Sprint-2 |
| **Administrator (Counselor)** | **Data Export** | USN-5 | As an admin, I want to view anonymized historical logs to track wellness progress over time. | Admin can access the stats view filtered by date. | Low | Sprint-2 |